DEWDRIVE · TRACEABILITY MATRIX

# DewDrive traceability matrix

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## DOCUMENT CONTROL

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| Document ID | DWD-RTM-001 | Version | 0.3 |
| Status | Draft | Date | 2026-10-25 |
| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/dewdrive | | |
| Source | docs/03b-traceability.md @ 4f86e9e | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.3 | 2026-10-25 | Amish Chadha | Generated by .kit/trace.py |
| 0.2 | 2026-10-25 | Amish Chadha | Generated by .kit/trace.py |
| 0.1 | 2026-10-25 | Amish Chadha | Generated by .kit/trace.py |

# DewDrive traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 12 user requirements, 14 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

| User requirement | Product requirement | Title | Evidence | Verification | Status |
|---|---|---|---|---|---|
| UR-001 | PR-001 | Water at the design point | DWD-CAL-001, DWD-BLD-001, DWD-STM-001, DWD-OPT-001 | Analysis (DWD-CAL-001 cyclic balance), TRL 3; field test at TRL 4 | Not met: 0.44 (C3); 0.37 to 0.41 under the sensitivities (C9, C12, C13) |
| UR-002, UR-001 | PR-002 | Water in dry nights | DWD-CAL-001, DWD-BLD-001 | Analysis (DWD-CAL-001), TRL 3; field test at TRL 4 | Met on paper: 0.31 (C4) |
| UR-008 | PR-003 | Sun-only regeneration | DWD-CAL-001, DWD-BLD-001 | Analysis (collector balance), TRL 3; logged at TRL 4 | Met on paper: 105 °C at noon (C2) |
| UR-008 | PR-004 | Own electrical loads | DWD-CAL-001, DWD-BLD-001 | Analysis (energy budget), TRL 3; measurement at TRL 4 | Met on paper: 23.6 against 35 Wh/day; 2.6 days (F1, F2) |
| UR-001 | PR-005 | Condenser rise | DWD-CAL-001, DWD-STM-001 | Analysis (condenser balance, painted underside D18), TRL 3; logged at TRL 4 | At risk: 11.9 K on the correlation (C2); about 14 to 16 K if the fins under the floor shed less heat (independent review) |
| UR-004 | PR-006 | Water quality | DWD-CAL-001, DWD-BLD-001 | Inspection (materials review), TRL 3; laboratory water test at TRL 4 | To verify at TRL 4; materials chosen to meet it |
| UR-012, UR-004 | PR-007 | Salt containment | DWD-CAL-001, DWD-BLD-001 | Analysis (pore and sump volumes, DWD-CAL-001 E), TRL 3; chamber test at TRL 4 | Met on paper: 47 % after one night, 54 % after three with the fan rule; sumps 0.58 L (E2 to E6) |
| UR-003 | PR-008 | Two actions per day | DWD-CAL-001, DWD-BLD-001 | Analysis (task analysis), TRL 3; co-design sessions at TRL 4 | Met by design |
| UR-007 | PR-009 | Portable | DWD-CAL-001, DWD-BLD-001 | Analysis (mass from model components), TRL 3; trial at TRL 4 | Met on paper: 23.4 kg; 36.1 kg with everything inside; 61.1 kg in all (G1, G2) |
| UR-006 | PR-010 | Survive the site | DWD-CAL-001, DWD-BLD-001 | Analysis (wind load), TRL 3; supplier data and cycling test at TRL 4 | To verify at TRL 4: two anchors of 234 N each at 25 m/s, specified 250 N, or 96 kg of ballast (H2 to H4) |
| UR-010 | PR-011 | Cost | DWD-CAL-001 | Analysis (priced BOM), TRL 3 | Over target by $175: $695 (J1) |
| UR-011 | PR-012 | Record performance | DWD-CAL-001, DWD-BLD-001 | Analysis (storage estimate), TRL 3; demonstration at TRL 4 | Met on paper: 8,640 records, 0.41 MB (I1) |

| User requirement | Product requirement | Title | Evidence | Verification | Status |
|---|---|---|---|---|---|
| UR-005 | PR-013 | Protect users | DWD-CAL-001, DWD-BLD-001, DWD-STM-001 | Analysis (surface temperatures), TRL 3; logged at TRL 4 | Met on paper: glazing outer skin 55 °C, fins 47 °C; 12.8 V DC (D1 to D3) |
| UR-009 | PR-014 | Local build and repair | DWD-CAL-001, DWD-BLD-001 | Inspection (build plan DWD-BLD-001 and BOM), TRL 3; build at TRL 4 | Met by design |

## Gaps

None.